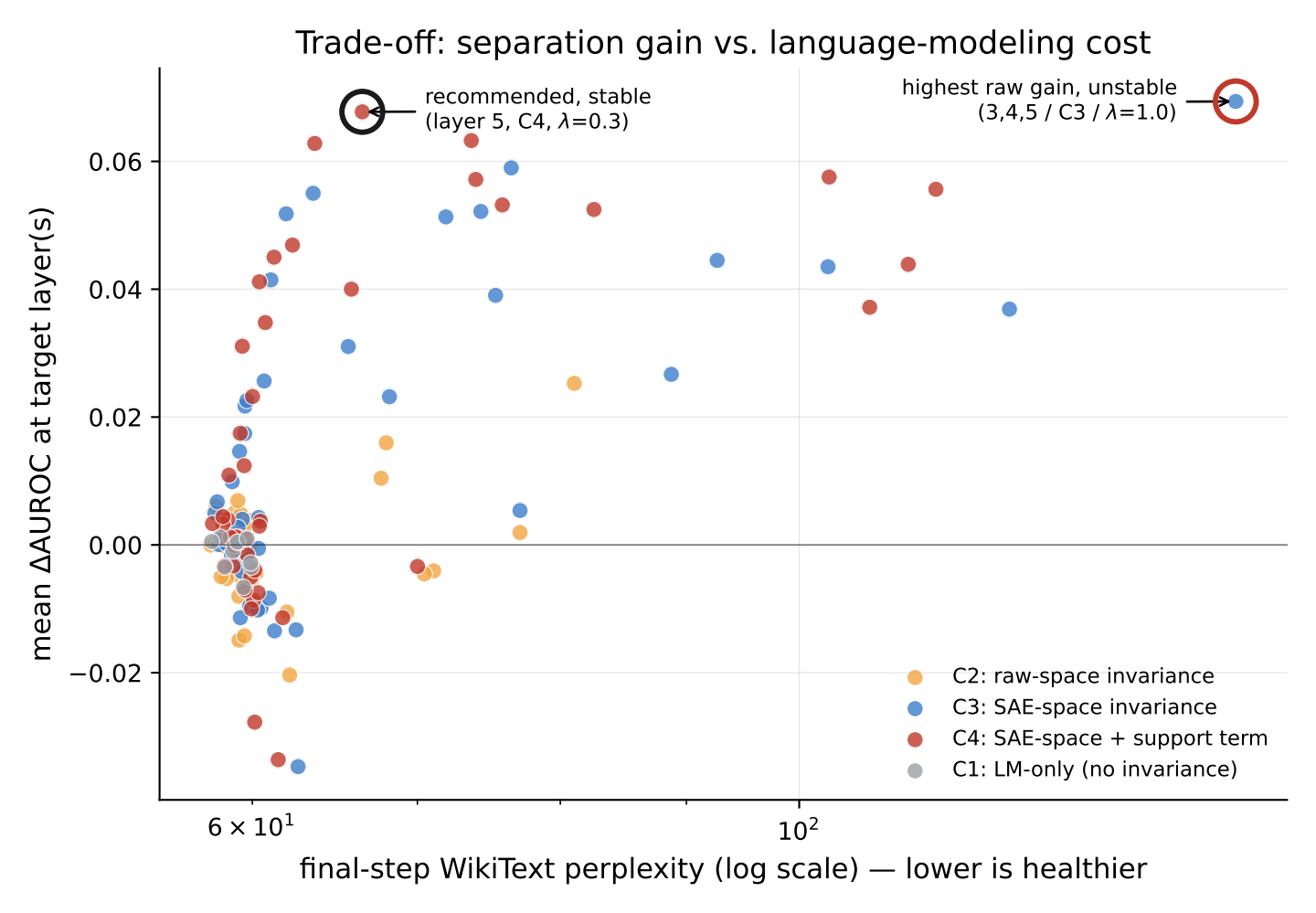
Trade-off: separation gain vs. language-modeling cost
recommended, stable
(layer 5, C4, λ=0.3)
highest raw gain, unstable
(3,4,5 / C3 / λ=1.0)
mean ΔAUROC at target layer(s)
0.06
0.04
0.02
0.00
−0.02
6 × 10^1
10^2
final-step WikiText perplexity (log scale) — lower is healthier
C2: raw-space invariance
C3: SAE-space invariance
C4: SAE-space + support term
C1: LM-only (no invariance)